Q1: cell-level time domain simulations, 2RC ECM

.The single cell model is exercised under three 1C profiles, step at 5 s, square with 10 s period, and sine with 20 s period.

$$s' = (1 - 2\epsilon)\, s + \epsilon \qquad \epsilon = 0.175$$

$$\mathrm{OCV}(s) = k_0 + \frac{k_1}{s'} + \frac{k_2}{s'^2} + \frac{k_3}{s'^3} + \frac{k_4}{s'^4} + k_5 s' + k_6 \ln(s') + k_7 \ln(1 - s')$$

$$v[k] = \mathrm{OCV}(s[k]) - R_0 I_p[k] - v_1[k] - v_2[k]$$

$$a_1 = e^{-\Delta t/(R_1 C_1)}, a_2 = e^{-\Delta t/(R_2 C_2)}$$

$$v_1[k+1] = a_1 v_1[k] + R_1 (1 - a_1) I_p[k]$$

$$v_2[k+1] = a_2 v_2[k] + R_2 (1 - a_2) I_p[k]$$

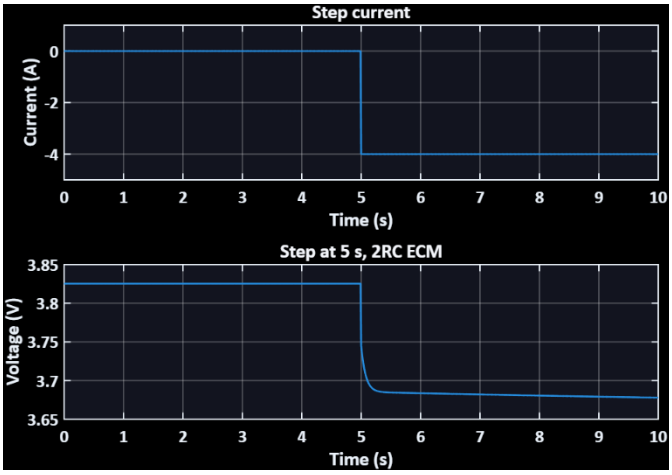

Figure 1:step current and voltage

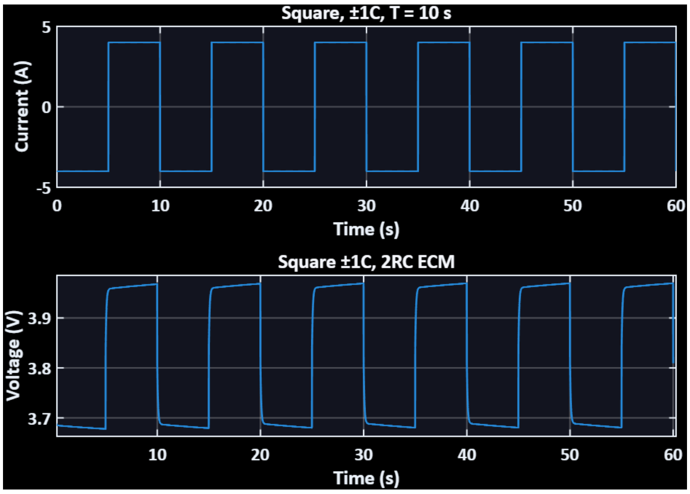

Figure 2: square current and voltage

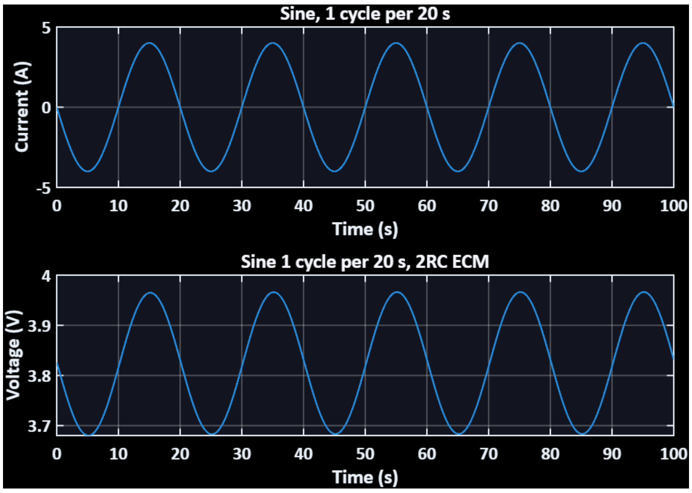

Figure 3: sine current and voltage

Q2, Nyquist, cell level

Shown in Figure 4, Frequency sweep from 1e-3 Hz to 1e3 Hz: This provides a semicircle with a trailing low-frequency tail. The high frequency intercept corresponds to the resistive component at approximately 0.020 ohms. The low-frequency intercept tends to the sum of R0, R1, and R2 as expected from the previous observations in the time domain. The locations of the arc tops are related to the corner frequencies of 1 divided by 2πR1C1 and 1 divided by 2πR2C2. This indicates the existence of both fast and slow polarization relaxation rates.

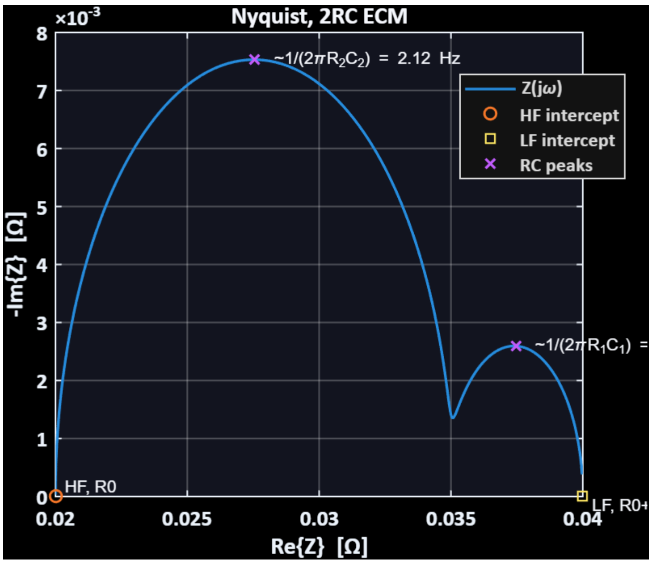

Figure 4: Nyquist plot with HF and LF

Q3, pack scaling, 110S72P, R int model

A single cell is scaled to a pack with 110 series cells and 72 parallel cells, using a reduced-order R int representation for the drive cycles. Here again, series impacts voltage and resistance. Now, when we go to the 110 series cells and 72 parallel cells pack configuration.

$$Q_{\text{pack}} = N_p\, C_{\text{cell}} = 72 \times 4.0\ \text{Ah} = 288\ \text{Ah}$$

$$R_{\text{pack}} = \frac{N_s}{N_p}\, R_0 = \frac{110}{72} \times 0.020\ \approx 0.03056\ \Omega$$

Q4, pack level drive cycles, UDDS, and HWFET

Applying UDDS and HWFET to the 110S72P R int pack starting from 50 percent SOC. For each cycle, there is a plot of current, pack terminal voltage, and SOC.

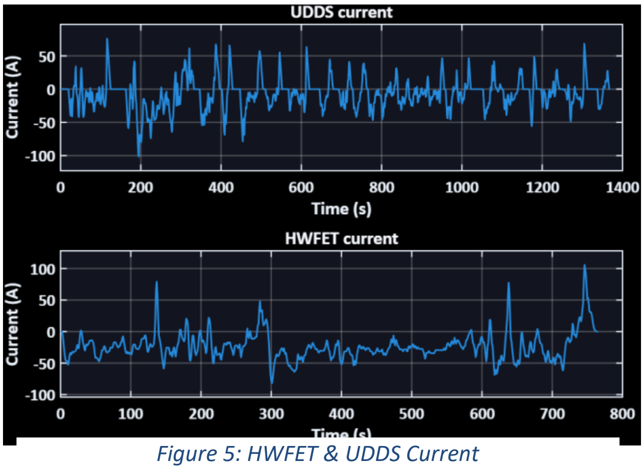

Figure 5: HWFET & UDDS Current

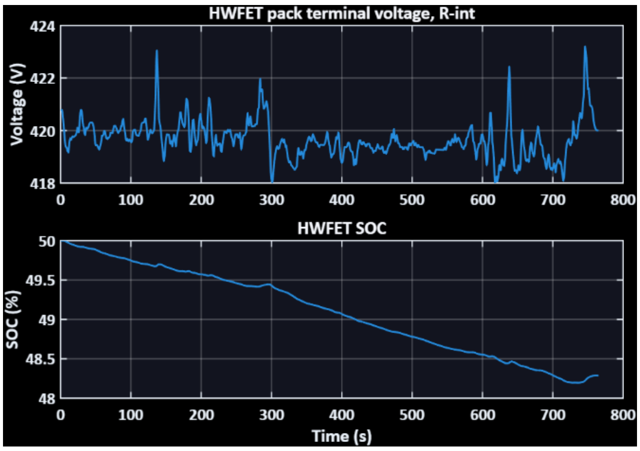

Figure 6: HWFET current, voltage, SOC

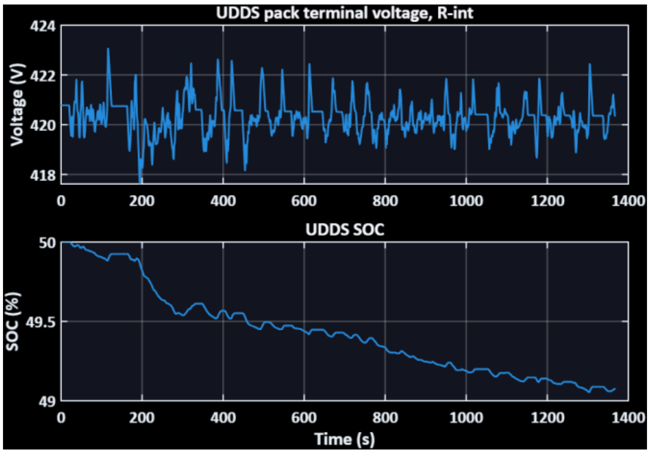

Figure 7: UDDS current, voltage, SOC

UDDS net Ah = -2.664 Ah, SOC end = 49.08 %
HWFET net Ah = -4.929 Ah, SOC end = 48.29 %

SOC decreases with the time integral of discharge current, so the cycle with greater net discharge content finishes with lower SOC. Larger magnitude currents also create deeper instantaneous voltage dips because the ohmic term scales with current magnitude, while the OCV function sets the operating band and the average level of the voltage trace.